

References

1. Gaynes BN, Lux L, Gartlehner G, Asher G, Forman-Hoffman V, Green J, et al. Defining treatment-resistant depression. *Depress Anxiety*. 2020;37(2):134-145. doi:10.1002/da.22968.
2. Rush AJ, Trivedi MH, Wisniewski SR, Nierenberg AA, Stewart JW, Warden D, et al. Acute and longer-term outcomes in depressed outpatients requiring one or several treatment steps: a STAR*D report. *Am J Psychiatry*. 2006;163(11):1905-1917. doi:10.1176/ajp.2006.163.11.1905.
3. Al-Harbi KS. Treatment-resistant depression: therapeutic trends, challenges, and future directions. *Patient Prefer Adherence*. 2012;6:369-388. doi:10.2147/PPA.S29716.
4. Perlis RH. A clinical risk stratification tool for predicting treatment resistance in major depressive disorder. *Biol Psychiatry*. 2013;74(1):7-14. doi:10.1016/j.biopsych.2012.12.007.
5. Kautzky A, Baldinger-Melich P, Kranz GS, Vanicek T, Souery D, Montgomery S, et al. A new prediction model for evaluating treatment-resistant depression. *J Clin Psychiatry*. 2017;78(2):215-222. doi:10.4088/JCP.15m10381.
6. Sheu YH, Magdamo C, Miller M, Das S, Blacker D, Smoller JW. AI-assisted prediction of differential response to antidepressant classes using electronic health records. *npj Digit Med*. 2023;6:73. doi:10.1038/s41746-023-00817-8.
7. Chekroud AM, Zotti RJ, Shehzad Z, Gueorguieva R, Johnson MK, Trivedi MH, et al. Cross-trial prediction of treatment outcome in depression: a machine learning approach. *Lancet Psychiatry*. 2016;3(3):243-250. doi:10.1016/S2215-0366(15)00471-X.
8. Chekroud AM, Bondar J, Delgadillo J, Doherty G, Wasil A, Fokkema M, et al. The promise of machine learning in predicting treatment outcomes in psychiatry. *World Psychiatry*. 2021;20(2):154-170. doi:10.1002/wps.20882.
9. Cepeda MS, Reps J, Fife D, Blacketer C, Stang P, Ryan P. Finding treatment-resistant depression in real-world data: how a data-driven approach compares with expert-based heuristics. *Depress Anxiety*. 2018;35(3):220-228. doi:10.1002/da.22705.
10. Fabbri C, Hagenaars SP, John C, Williams AT, Shrine N, Moles L, et al. Genetic and clinical characteristics of treatment-resistant depression using primary care records in two UK cohorts. *Mol Psychiatry*. 2021;26(7):3363-3373. doi:10.1038/s41380-021-01062-9.

11. Iveson MH, Ball EL, Lo CWH, Falis M, Lewis CM, Whalley HC. Treatment resistant depression in electronic health records: definitions matter. *BMC Psychiatry*. 2026;26(1):453. doi:10.1186/s12888-026-08085-y.
12. Liberman JN, Davis T, Pesa J, Chow W, Verbanac J, Heverly-Fitt S, et al. Predicting incident treatment-resistant depression: a model designed for health systems of care. *J Manag Care Spec Pharm*. 2020;26(8):987-995. doi:10.18553/jmcp.2020.26.8.987.
13. Lage I, McCoy TH, Perlis RH, Doshi-Velez F. Efficiently identifying individuals at high risk for treatment resistance in major depressive disorder using electronic health records. *J Affect Disord*. 2022;306:254-259. doi:10.1016/j.jad.2022.02.046.
14. Walsh CG, Ripperger M, McCoy TH, Castro V, Hu Y, Kirchner HL, et al. Generalizability of risk models for treatment-resistant depression across three health systems. *medRxiv*. 2025. Preprint. doi:10.1101/2025.05.21.25328089.
15. Lee DY, Kim N, Park C, Gan S, Son SJ, Park RW, et al. Explainable multimodal prediction of treatment-resistance in patients with depression leveraging brain morphometry and natural language processing. *Psychiatry Res*. 2024;334:115817. doi:10.1016/j.psychres.2024.115817.
16. Hegselmann S, von Arnim G, Rheude T, Kronenberg N, Sontag D, Hindricks G, et al. Large language models are powerful electronic health record encoders. *arXiv:2502.17403*. 2025.
17. Shmatko A, Jung AW, Gaurav K, Brunak S, Mortensen LH, Birney E, et al. Learning the natural history of human disease with generative transformers. *Nature*. 2025;647(8088):248-256. doi:10.1038/s41586-025-09529-3.
18. Waxler S, Blazek P, White D, Sneider D, Chung K, Nagarathnam M, et al. Generative medical event models improve with scale. *arXiv*. 2025;arXiv:2508.12104. doi:10.48550/arXiv.2508.12104.
19. Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*. 2024;385:e078378. doi:10.1136/bmj-2023-078378.
20. Forthman KL, Kuplicki R, Thompson WK, Nemeroff CB, Si Y, Fan CC, et al. Treatment resistant depression: socio-demographic characteristics, comorbidity and treatment patterns from the All of Us Research Program. *J Affect Disord*. 2025;390:119858. doi:10.1016/j.jad.2025.119858.

21. Hegselmann S, Shen SZ, Gierse F, Agrawal M, Sontag D, Jiang X. A data-centric approach to generate faithful and high quality patient summaries with large language models. *Proc Mach Learn Res.* 2024;248:339-379.
22. Reimers N, Gurevych I. Sentence-BERT: sentence embeddings using Siamese BERT-networks. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*; 2019 Nov; Hong Kong. Stroudsburg (PA): Association for Computational Linguistics; 2019. p. 3982-3992. doi:10.18653/v1/D19-1410.
23. Xiao S, Liu Z, Zhang P, Muennighoff N, Lian D, Nie JY. C-Pack: packed resources for general Chinese embeddings. In: *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '24)*; 2024 Jul 14-18; Washington (DC). New York: ACM; 2024. p. 641-649. doi:10.1145/3626772.3657878.
24. Li C, Qin M, Xiao S, Chen J, Luo K, Shao Y, et al. Making text embedders few-shot learners. *arXiv:2409.15700*. 2024.
25. Zhang Y, Li M, Long D, Zhang X, Lin H, Yang B, et al. Qwen3 embedding: advancing text embedding and reranking through foundation models. *arXiv:2506.05176*. 2025.
26. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: machine learning in Python. *J Mach Learn Res.* 2011;12:2825-2830.
27. Chen T, Guestrin C. XGBoost: a scalable tree boosting system. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16)*; 2016 Aug 13-17; San Francisco (CA). New York: ACM; 2016. p. 785-794. doi:10.1145/2939672.2939785.
28. González HM, Vega WA, Williams DR, Tarraf W, West BT, Neighbors HW. Depression care in the United States: too little for too few. *Arch Gen Psychiatry.* 2010;67(1):37-46. doi:10.1001/archgenpsychiatry.2009.168.
29. Alegría M, Chatterji P, Wells K, Cao Z, Chen CN, Takeuchi D, et al. Disparity in depression treatment among racial and ethnic minority populations in the United States. *Psychiatr Serv.* 2008;59(11):1264-1272. doi:10.1176/ps.2008.59.11.1264.
30. Goldberger J, Roweis S, Hinton G, Salakhutdinov R. Neighbourhood components analysis. In: *Saul LK, Weiss Y, Bottou L, editors. Advances in Neural Information Processing Systems 17*

(NIPS 2004). Cambridge (MA): MIT Press; 2005. URL: <https://proceedings.neurips.cc/paper/2004/hash/42fe880812925e520249e808937738d2-Abstract.html>

31. Cawley GC, Talbot NLC. On over-fitting in model selection and subsequent selection bias in performance evaluation. J Mach Learn Res. 2010;11:2079-2107.